MIL-R-3533B
25 APRIL 1964
SUPERSEDING
MIL-R-3533A
13 SEPTEMBER 1960

MILITARY SPECIFICATION

RUBBER, SYNTHETIC; SHEET, STRIP AND MOLDED

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the requirements for sheet, strip and molded synthetic rubber for use as torpedo gasket material.

★ **1.2 Classification.** Synthetic rubber shall be of the following types and grades as specified (see 6.2).

Type I —O-Rings
Type II —Molded parts (other than O-Rings), sheet and strip
Grade A—Medium hard
Grade B—Medium soft

★ 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-515 —Plywood; Container Grade
UU-P-268 —Paper; Kraft, Untreated, Wrapping
UU-T-111 —Tape; Paper, Gummed (Sealing and Securing)
PPP-B-566—Boxes; Folding, Paperboard
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585—Boxes; Wood, Wirebound
PPP-B-591—Boxes; Fiberboard Wood Cleated
PPP-B-601—Boxes; Wood, Cleated Plywood
PPP-B-621—Boxes; Wood, Nailed and Lock Corner
PPP-B-636—Boxes; Fiber
PPP-B-676—Boxes; Set-up, Paperboard

MILITARY

MIL-P-116 —Preservation; Methods of
MIL-P-4861 —Packing, Preformed, Rubber, Packing, Packaging of
MIL-L-10547—Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

FEDERAL

Federal Test Method Standard No. 601 —Rubber: Sampling and Testing

MILITARY

MIL-STD-10 —Surface Roughness, Waviness and Lay
MIL-STD-129—Marking for Shipment and Storage
MIL-STD-413—Visual Inspection Guide for Rubber O-Rings

FSC 5330

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MS 33666 —Packing, Preformed-Aeronautical, Elastomeric, Range of Sizes

PUBLICATIONS

AIR FORCE-NAVY AERONAUTICAL BULLETIN

No. 438—Age Controls for Synthetic Rubber Parts

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

★ **2.2 Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM–D746–57T—Brittleness Temperature of Plastics and Elastomers by Impact.

ASTM–D1329–60—Evaluating Low-Temperature Characteristics of Rubber and Rubber-like Materials by a Temperature-Retraction Procedure (TR Test)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania).

3. REQUIREMENTS

★ **3.1 Material.** Synthetic rubber furnished under this specification shall be a vulcanized elastomeric compound which is resilient, oil and salt water resistant and shall not adhere to nor cause corrosion of metal parts to which it is assembled.

3.2 Forms. Synthetic rubber shall be furnished in sheet, strip, cut or molded form as specified (see 6.2).

3.2.1 *Sheet and strip.* Unless otherwise specified in the contract or order, the following dimensions and tolerances shall apply. The width of sheet material shall be 36 inches plus or minus 1 inch and the width of strip material as specified with tolerances of plus or minus 5 percent. The length shall have a tolerance of plus or minus 1 percent and tolerances on thickness shall be as shown in Table I.

★ TABLE I. *Tolerance in thickness*

Thickness – Inch	Tolerance – Inch (plus or minus)
Less than 0.0625	0.010
0.0625 to 0.125, inclusive	0.016
Over 0.125 to 0.500, inclusive	0.031
Over 0.500 to 1.000, inclusive	0.047
Over 1.000	0.063

3.2.2 *Cut or molded parts.* Shape, dimensions and tolerances shall be as specified on the drawing or in the contract or order.

3.3 Physical requirements. The synthetic rubber material shall conform to the physical requirements specified in Table II.

3.4 Performance.

3.4.1 *Crush test.* All O-Ring material when tested as specified in 4.5.4 shall be capable of withstanding high compressive loadings without cracking or otherwise disintegrating.

3.4.2 *Permanent set.* When tested for permanent set as specified in 4.5.5, the samples shall show no evidence of splitting or breaking.

3.4.3 *Low temperature flexibility.* Samples after being subjected to the test described in 4.5.7 shall show no visible signs of cracks, checks, or other damage.

3.4.4 *Adhesion to and corrosion of metal.* When tested in accordance with 4.5.8, the

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specimens shall be readily separated from the metal with which they have been in close contact and shall not cause corrosion pitting of such metal. A slight discoloration of the metallic contact surfaces will be acceptable.

3.5 Finish.

3.5.1 *General.*

3.5.1.1 *Finish and trimming.* Mold flash shall be removed in such a manner that the pieces conform to the requirements specified herein and on the applicable drawing or Military Standard.

★ 3.5.1.2 *Sealing surfaces.* Sealing surfaces shall be smooth and free from all defects, blemishes, or irregularities except as permitted on the applicable drawing or Military Standard. Inspection shall be visual and without magnification in accordance with MIL–STD–413. The entire surface of the O-Ring is considered as a sealing surface.

3.5.2 *O-Rings.* Any method used to trim mold flash from O-Rings shall not remove any of the original molded finish of the ring. Unless specifically stated otherwise, the cross sectional dimensions of the flash remaining after trimming shall not exceed 0.003 inch radially and 0.005 inch circumferentially (or laterally).

★ TABLE II. *Physical properties*

PROPERTY	TYPE I		TYPE II	
	Grade A	*Grade B*	*Grade A*	*Grade B*
Original				
Tensile Strength—psi (min)	1850	1650	2000	1800
Ultimate Elongation—% (min)	200	300	250	350
Tear Resistance—lbs/in thickness (min)	25	35	210	250
Tension Set—200% elongation—% (max)	..	..	12	
300% elongation—% (max)	..	..	..	15
Tensile Modulus—300% elongation—psi (min)	..	..	.	1000
Permanent Set—150 elongation—% (max)	2	2	..	..
Compression Set—25% deflection—% (max)	..	..	10	10
Crush Test—5600 lb load—% recovery (min)	90	90	..	..
Temperature Retraction—50% elongation and 10% recovery—°F (max)	—10	—20	..	..
Hardness—Shore "A" points	80±5	70±5	80±5	70±5
Air Aged — 168 ± 1 hr at 70 ± 2°C				
Tensile Strength — psi (min)	1750	1500	1900	1700
Ultimate Elongation — % (min)	175	275	225	325
Tension Set—200% elongation — % (max)	..	..	15	..
300% elongation — % (max)	..	..	..	18
Permanent Set — 150% elongation — % (max)	2	2	..	..
Crush Test — 5600 pound load — % recovery (min)	85	85	..	..
Hardness Change—Shore "A" points — (max)	±5	±5	±5	±5
Oil Aged — 168 ± 1 hr at 70 ± 2°C				
Tensile Strength — psi (min)	1700	1500	1900	1700
Ultimate Elongation — % (min)	175	275	225	325
Tension Set—200% elongation — % (max)	..	..	15	..
300% elongation — % (max)	..	..	..	20
Permanent Set — 150 elongation — % (max)	4	4	..	..
Crush Test — 5600 lb load — % recovery (min)	80	80	..	..
Temperature Retraction — 50% elongation and 10% retraction — °F (max)	+10	0	..	..
Volume Change — % (max)	±8	±8	±10	±10
Hardness Change — Shore "A" Points — (max)	±10	±10	±10	±10
Synthetic Sea Water Aged — 168 ± 1 hr at 70 ± 2°C				
Tensile Strength — psi (min)	1750	1500	1900	1700
Ultimate Elongation — % (min)	175	225	225	325
Crush Test — 5600 lb load — % recovery (min)	85	85	..	..
Volume Change — % (max)	±5	±5	±5	±5
Hardness Change — Shore "A" Points — (max)	±5	±5	±5	±5

Note: The symbol " .. " means that test is not applicable.

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3.5.3 Minor surface defects that do not affect serviceability of the material shall not be cause for rejection.

3.6 Age limitations.

3.6.1 Age limitations shall be in accordance with ANA Bulletin No. 438.

3.7 Identification.

★ 3.7.1 *Sheet and strip.* Unless otherwise specified in the contract or order, sheet material shall be marked to show the specification number, type and grade, date of manufacture (by calendar quarters) and the manufacturer's code identification and compound number. The identification shall appear in rows of constantly recurring symbols from one end of the sheet to the other, spaced approximately 5 inches apart. The manufacturer's code identification and compound number shall appear immediately below the specification number. The symbol shall be clearly legible and not less than ¼ inch high and shall be applied by using a marking fluid that is not deleterious to the rubber. The marking shall not be obliterated by normal handling nor by the action of lubricating oil.

3.7.2 *Molded shapes.* When the size of the product permits, the identifying symbol, as in 3.7.1, shall be marked once on each part.

3.7.3 *Small parts.* Where material dimensions are such that markings cannot be made in accordance with the above, markings shall appear on the container only and in accordance with 5.3.

3.8 Workmanship. Workmanship shall be of a quality to assure the materials comply with the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

★ **4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *Lot.* A lot shall consist of all the material of any one type and one grade produced from the same batch of rubber compound submitted at one time. A batch shall be the quantity of material processed through a mill or mixer during a single run and used to fabricate parts, sheets or strips.

★ 4.2.2 *Sampling for lot inspections.* From each lot a random sample shall be selected in accordance with Table I, Section 6, of Federal Test Method Standard No. 601.

★ 4.2.3 *Sampling for quality conformance testing.* The manufacturer shall furnish the necessary samples molded from the same material and under the same conditions as that used in the finished product. When Type I material is specified on the order, the testing samples shall consist of eighteen O-Rings (Size No. 214 of MS33666) and twelve compression molded discs 0.250±0.005 inch thick and not less than one inch in diameter. When Type II material is specified, the samples shall be four 6 by 6 by 0.075 ±0.005 inch platen sheets, three compression set discs 0.500 ±0.010 inch thick and 1.129 ±0.005 inch in diameter, and twelve compression molded discs 0.250 ±0.005 inch thick and not less than one inch in diameter.

4.3 Inspection.

4.3.1 *Dimensions, finish, marking and workmanship.* Each unit of the sample selected in accordance with 4.2.2 shall be carefully examined to insure conformance with respect to dimensions, finish, marking and workmanship. The number of nonconforming units shall be recorded and the lot rejected without testing if the number of defective units in the sample equals or exceeds the rejection number specified in Table I, Section 6, of Federal Test Method Standard No. 601.

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★ 4.3.2 *Packaging and packing.* The finished product shall be packaged, marked, and prepared for shipment in compliance with Section 5 of this specification. Inspection shall consist of the Group A tests listed in Section 4 of MIL–P–116.

4.4 Classification of tests.

★ 4.4.1 *Quality conformance tests.* Quality conformance tests for acceptance shall consist of those tests specified in Table III which are applicable to the type material submitted and for the "as received" condition only.

★ 4.4.2 *Rejection.* Failure of any sample of synthetic rubber to conform to one or more of the test requirements of this specification shall be cause for rejection of the lot represented.

★ **4.5 Test methods.** Unless otherwise specified, the physical properties listed in Table III shall be determined in accordance with Federal Test Method Standard No. 601.

★ 4.5.1 Hardness of the compound shall be determined as specified in Table III except that test samples shall be three compression molded discs 0.250 ±0.005 inch thick and not less than one inch in diameter. Hardness checks shall not be made on actual gaskets.

4.5.2 Tests specified for O-Rings (Type I material) shall be performed on the actual samples of rings submitted for acceptance testing, when possible.

4.5.3 Tear resistance shall be measured in two planes on separate samples; one with the mold seam and the second at right angles to the mold seam. A minimum of two samples shall be tested. In conducting the test, the O-Ring shall first be severed to open the ring. A cut shall then be made through the center of the cross sectional diameter of one end. Lengthwise, this cut shall be about 1¼ inches, sufficient to allow the two halves formed to be inserted into the jaws of a testing machine (Scott tester, or equivalent). The tear resistance shall be calculated as follows:

$$\text{Tear resistance, pounds per inch of thickness} = \frac{F}{T}$$

where: F = minimum tearing force in lbs. and

T = the thickness of the specimen in inches.

★ 4.5.4 *Crush test (Type I only).* Three O-Rings shall be selected at random for testing. They shall be twisted (inside out) one-half turn about the central axis of the cross section and then placed in separate locations between two rigid steel plates having a surface roughness of 32 to 63 micro inches rms. (in accordance with Standard

★ TABLE III. *Quality conformance tests*

Property	*Applicable type*	*Method (FED-STD-601 or Paragraph no.)*
a. Hardness	Types I and II	3021
b. Tensile Strength	Types I and II	4111
c. Ultimate Elongation	Types I and II	4121
d. Tensile Stress	Types I and II	4131
e. Tear Resistance	Type II	4211
	Type I	see 4.5.3
f. Compression Set	Type II	3311
g. Crush Test	Type I	see 4.5.4
h. Tension Set	Type II	4411
i. Permanent Set	Type I	see 4.5.5
j. Temperature Retraction	Type I	see 4.5.6
k. Low Temperature Flexibility	Type II	see 4.5.7
l. Adhesion & Corrosion	Type II	see 4.5.8
m. Volume Change	Types I and II	see 4.5.10.3

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MIL–STD–10). An axial compression load of 16,800 pounds (or 5600 pounds for each ring in the layer) shall be applied. The plates shall be secured to maintain this load for 20 hours by a suitable means (mechanical fixture). The rings shall then be removed and inspected immediately for cracking or disintegration. Following this inspection, the rings shall be stored at room temperature in the free condition and the percent recovery of the ring cross section determined at the end of six hours. No cracking or disintegration shall be evident and the cross sectional dimensions of the O-Rings shall have attained the values specified in Table II in respect to the "as manufactured" size at the thinnest section.

★ 4.5.5 *Permanet set (Type I only).* Permanent set on three O-Rings shall be made by measuring the ID and then stretching over a tapered cone (lubricated with ASTM No. 1 oil) until the ID of the seal has been stretched 150% (if the ID measures 0.984 inch, the ring shall be stretched to 2.460 inches ID). The ring shall be held in this stretched state for ten minutes, removed from the cone, and the ID determined after ten minutes rest. Permanent set shall be calculated from the as-measured ID before and after the test. Any breaking or cracking of the test samples shall be cause for rejection.

★ 4.5.6 *Temperature retraction (Type I only).* This test shall be in accordance with ASTM D1329–60 except that O-Ring samples (Size No. 214 of MS33666) shall be used and three samples tested. After installation, samples shall be elongated 50% as determined by the rod markings and the rate of warming shall be 0.6°C(1°F) per minute. The temperature retraction 50/10 reading of these seals shall be determined and recorded.

★ 4.5.7 *Low temperature flexibility (Type II only).* This test shall be in accordance with ASTM D746–57T except that the thickness of the samples shall be 0.075 ± 0.005 inch. The test temperature for exposure of the samples shall be —17 ±2°C (1.4 ±4°F).

4.5.8 *Adhesion and corrosion.* Specimens shall be tested in accordance with Method 14211 of Federal Test Method Standard No. 601. For Type I material, the actual O-Ring shall be used for the test specimen in place of the one inch square specimen.

4.5.9 *Air aging.* Air aging shall be conducted in accordance with Federal Test Method Standard No. 601, Method 7221.

★ 4.5.10 *Aging in liquids.* Specimens shall be aged in accordance with Method 6001 of Federal Test Method Standard No. 601 using Medium No. 1 for the oil and synthetic sea water for the water medium. Tests shall be conducted for a period of 168 ±1 hour at a temperautre of 70 ±2°C (158 ±4°F).

4.5.10.1 *Synthetic sea water.* The synthetic sea water used in the aging tests shall be formulated as follows:

Sodium Chloride	23 gr
Sodium Sulfate Decahydrate	8 gr
Stock Solution	20 ml
Distilled Water	To make 1 liter

4.5.10.1.1 *Stock solution.* The stock solution used in formulating the synthetic sea water shall have the following composition:

Potassium Chloride	10 gr
Potassium Bromide	45 gr
Magnesium Chloride Sexahydrate	550 gr
Calcium Chloride Sexahydrate	110 gr
Distilled Water	To make 1 liter

4.5.10.2 Tensile strength and ultimate elongation property determinations after liquid aging shall be performed in accordance with Method 6121 of Federal Test Method Standard No. 601 except that a recovery period of twenty hours in the original test liquids will be allowed.

★ 4.5.10.3 Changes in volume after liquid aging shall be performed in accordance with Method 6211 of Federal Test Method Standard No. 601. O-Ring samples shall be used

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for Type I material and compression molded discs 0.250 ±0.005 inch thick and not less than one inch in diameter shall be used for Type II material. The discs used for hardness tests may also be used for volume change determinations. Tests shall be run in triplicate.

5. PREPARATION FOR DELIVERY

★ **5.1 Packaging.** Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Type I material.* Type I material (O-Rings) shall be packaged in accordance with Specification MIL–P–4861.

5.1.1.2 *Type II material.*

5.1.1.2.1 *Sheets.* Rubber sheets shall be interleaved with paper (Specification UU–P–268) which will extend over the full area of contact between the sheets. Unit quantity shall be a stack not to exceed ten inches in height.

5.1.1.2.2 *Strip.* Rubber strips shall be wound on suitable cores which will provide rigid support and which will not distort nor change shape during handling or shipping. Each roll shall be wrapped in Kraft paper conforming to Specification UU–P–268 and sealed with tape conforming to Specification UU–T–111.

5.1.1.2.3 *Molded parts.* Molded rubber parts shall be packaged in accordance with Method III of Specification MIL–P–116. Unit and intermediate containers when used shall conform to Specifications PPP–B–566, PPP–B–636, and PPP–B–676. The gross weight of the unit and intermediate container shall be subject to the limitation of the box specification. Sealing and closures shall conform to the applicable box specification.

5.1.2 *Level C.* Synthetic rubber shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 Type I material. Type I material shall be packed in accordance with Specification MIL–P–4861.

★ 5.2.1.2 *Type II material.* Shipping containers shall contain identical synthetic rubber items of the same composition and size and shall enclose the contents in a snug, tight-fitting manner. For sheet, packaged as specified in 5.1.1.2.1, the outside height of containers shall not exceed 10 inches. Rubber strip, packaged as specified in 5.1.1.2.2, shall be packed one roll per container. Unless specified by the procuring activity, Type II material shall be packed in export-type wooden containers conforming to Specification PPP–B–601 or PPP–B–621. Plywood, if used, shall conform to Specification NN–P–515, Type I or II, Class 2. Shipping containers shall have case liners conforming to Specification MIL–L–10547 and appendix thereto. The gross weight of the container shall not exceed 200 pounds.

★ 5.2.2 *Level B.* Material, Types I and II, preserved and packaged in accordance with 5.1.1 shall be packed in domestic type exterior containers meeting the requirements of Specification PPP–B–585, PPP–B–591, PPP–B–601, PPP–B–621, PPP–B–636 or PPP–B–576, respectively. Exterior containers shall be of minimum cube and tare consistent with the protection required. When practical, each container shall contain an identical number of items and shall be of uniform size. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds, that of fiberboard boxes shall not exceed the weight limitations of the applicable specifications.

5.2.3 *Level C.* Packages which require overpacking for acceptance by common carrier shall be packed in exterior type shipping containers in a manner which will insure safe delivery and acceptance at point of delivery. Containers shall comply with the Consolidated Freight Classification Rules or other regulations as applicable to the mode of transportation.

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5.3 Marking. Unit packages and intermediate and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129. The nomenclature shall be the exact nomenclature for the part or item that has been approved under the contract or order.

5.3.1 In addition to any special marking required by the contract or order, the following information shall be added:

Date of Cure (Calendar quarter) (Year)
STORE IN COOL, DRY PLACE.

6. NOTES

6.1 Intended use. Synthetic rubber covered by this specification is intended for all types of synthetic rubber gaskets and seals for torpedo use.

★ **6.1.1** The grades of this specification corresponding to the classes of the previous issue are as follows:

Current Issue	*Previous Issue*
Grade A	Class 1
Grade B	Class 2

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and grade of material required (see 1.2).
(c) Form of material required (see 3.2).
(d) Dimensions required (see 3.2.1 and 3.2.2).
(e) Shape required (see 3.2.2).
(f) Tolerance for cut or molded material (see 3.2.2).
(g) Marking required (see 3.7).
(h) Selection of applicable level of preservation, packaging, and packing (see 5.1 and 5.2).
(i) Special packing, marking, if required (see 5.3.1).

★ **6.3 Transportation description.**

★ **6.3.1** Transportation descriptions and minimum weights applicable to this commodity are:

Rail: Carload minimum weight 30,000 lbs.
Motor: Truckload minimum weight 30,000 lbs.

★ **6.4 Changes from previous issue.** The outside margins of this document have been marked "★" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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★ **Review Activities:**
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★ **Preparing Actviity:**
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